

FINAL PROJECT PRESENTATION

Nazsya AI Tutor

AI Smart Tutor Using Advanced Prompt Engineering Techniques

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STMIK Kaputama Binjai · Google Prompt Engineering Course · 2026

Structured Prompting

Multi-Level Explanation

Quiz Generator

AI Evaluator

Background & Problem Statement

Why This Project?

AI is growing fast in education — it can explain topics, give quizzes, create study plans, and give feedback.

But without proper prompt design, AI responses tend to be too broad, inconsistent, or not suited for different types of learners.

Problems Identified:

AI responses often lack structure for educational use

Learners need explanations at different levels

Students need study plans, quizzes & feedback too

AI evaluation needs clear rubric criteria

Prompt techniques need testing & comparison

 Solution: Nazsya AI Tutor — an AI-powered educational assistant using advanced prompt engineering techniques

Project Objectives & Scope

1

Build an AI tutor app using Streamlit + LLM API

2

Use structured prompting for organized educational responses

3

Provide multi-level explanations: Beginner, Intermediate, Advanced

4

Generate personalized study plans by study duration

5

Create interactive quizzes for self-assessment

6

Build an AI Answer Evaluator using rubric + role-based prompting

7

Compare basic vs optimized prompts to measure impact

 Scope: Web-based app (Streamlit). Started with Gemini API, migrated to OpenRouter + DeepSeek Chat for stability.

Tools & Technologies



Python

Main programming language



Streamlit

Web UI framework



OpenRouter API

Final LLM provider



DeepSeek Chat

Final AI model used



Gemini API

Initial prototype LLM



.env Config

Secure API key management

⚡ API Migration: Gemini (prototype) → OpenRouter + DeepSeek Chat (final) — for better stability & flexibility

System Overview & Workflow

1

User Input

Enter learning topic, choose level, duration & evaluator role

2

Prompt Engineering

System applies structured, few-shot, multi-level prompting

3

AI Response

OpenRouter + DeepSeek Chat generates educational content

4

Output Display

Shows explanation, study plan, quiz, and evaluation

9. Prompt engineering analysis
10. Performance evaluation table

Overall, Nazsya AI Tutor demonstrates how prompt engineering techniques can improve the quality, structure, consistency, and educational usefulness of AI-generated learning content.

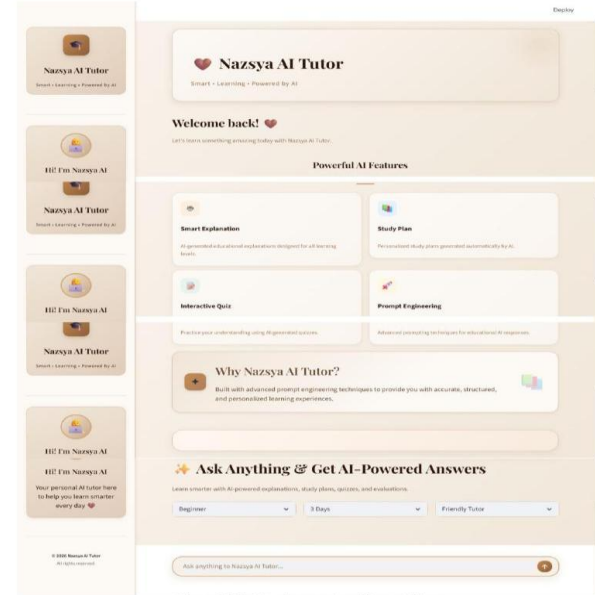


Figure 6. 1 Final interface overview of Nazsya AI Tutor

Prompt Engineering Techniques Used

Structured Prompting

Organizes AI responses into clear sections: definition, explanation, examples, and summary

Few-Shot Prompting

Provides example-based inputs so the AI learns the expected response format

Rubric-Based Evaluation

Evaluates student answers using Accuracy (0-5), Completeness (0-5), and Clarity (0-5)

Multi-Level Explanation

Adapts the explanation style to Beginner, Intermediate, or Advanced level learners

Role-Based Prompting

AI acts as Friendly Tutor, Strict Lecturer, or Professional Examiner based on user selection

Prompt Optimization

Compares basic vs optimized prompts and measures clarity, structure, and consistency

Main Features



Smart Explanation

- Structured AI-generated explanations
- 3 learner levels: Beginner, Intermediate, Advanced
- Covers definition, examples & summary



Study Plan Generator

- Personalized plan: 3, 5, or 7 Days
- Day-by-day learning structure
- Matches the learner's selected level



Interactive Quiz

- Auto-generated multiple-choice questions
- QUIZ_START / QUIZ_END parsing format
- Immediate score feedback after submission



AI Answer Evaluator

- Rubric-based scoring (0–5 per criterion)
- Role: Friendly Tutor / Strict Lecturer / Examiner
- Strengths, weaknesses & improvement tips

Interactive Quiz Feature



Figure 3. 13 Interactive quiz display in the final Nazsya AI Tutor application



Figure 3. 14 Quiz score result after user submission

Aspect	Initial Quiz Generator	Final Interactive Quiz Generator
API Provider	Gemini API	OpenRouter API
Model	Gemini 2.5 Flash Lite	DeepSeek Chat
Quiz Format	Text-based quiz with answer keys	Structured quiz using QUIZ_START and QUIZ_END
User Interaction	User reads quiz manually	User answers using radio buttons

How It Works

- 1 User selects topic and learner level
- 2 AI generates quiz using QUIZ_START / QUIZ_END format
- 3 App parses quiz into interactive radio buttons
- 4 User selects answers and clicks Submit Quiz
- 5 App shows score (e.g. 2/3) and feedback instantly

Result: Users get instant quiz score + motivational feedback

AI Answer Evaluator

Rubric-Based Scoring

Accuracy

Is the answer factually correct?

Score: 0–5

Completeness

Does it cover the key concepts?

Score: 0–5

Clarity

Is it easy to read and understand?

Score: 0–5

Evaluator Roles

Friendly Tutor
Warm, encouraging,
supportive

Strict Lecturer
Critical, firm, academic
tone

Professional Examiner
Formal, objective, rubric-
based

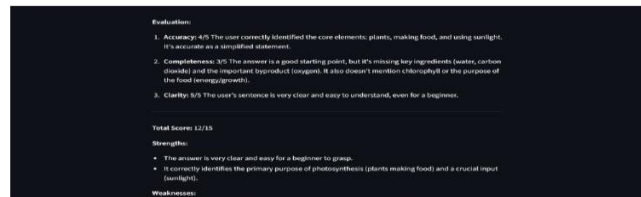


Figure 4.11 AI-generated rubric-based evaluation using Accuracy, Completeness, and Clarity criteria

Summary of AI-generated rubric-based evaluation:

The rubric-based evaluator assessed the user's answer: "Photosynthesis is the process plants use to make food using sunlight." The AI evaluated the answer using three criteria: Accuracy, Completeness, and Clarity.

The answer received 4/5 for Accuracy because it correctly mentioned plants, food production, and sunlight. It received 3/5 for Completeness because it missed important components such as water, carbon dioxide, oxygen, and chlorophyll. It received 5/5 for Clarity because the sentence was simple and easy to understand.

The total score was 12/15. The evaluator also provided strengths, weaknesses, and suggestions for improvement, showing that rubric-based prompting can produce more structured and objective feedback.

4.2.2 Final Rubric-Based Evaluator Implementation in OpenRouter Version

After the basic answer evaluator was implemented, the evaluation prompt was refined using rubric-based prompting in the final OpenRouter-based version of Nazsya AI Tutor.

In this implementation, the evaluator assessed the student's answer using three criteria: Accuracy, Completeness, and Clarity. Each criterion was scored from 0 to 5, and the final total score was calculated out of 15.

This rubric-based approach improved the evaluator because the feedback became more specific, measurable, and educationally meaningful.

Basic vs Optimized Prompt

✗ Basic Prompt

- Generic, one-size-fits-all answer
- No clear structure or sections
- Doesn't adapt to learner level
- Moderate clarity and consistency
- No study plan or quiz included



☑ Optimized Prompt

- Structured: definition, explanation, examples
- Summary + real-world analogy included
- Adapts to Beginner / Intermediate / Advanced
- Very high clarity and consistency
- Includes study plan + interactive quiz

Key insight: Optimized prompting significantly improves structure, personalization, and educational value of AI responses

Testing Results

Feature Tested	Testing Focus	Result
Structured Answer Format	Response organization using predefined sections	Successful
Beginner-Level Explanation	Simple, friendly, and analogy-based explanation	Successful
Intermediate-Level Explanation	More detailed explanation with scientific terminology	Successful
Advanced-Level Explanation	Technical and academic explanation style	Successful
Initial Gemini API Integration	Prototype testing and early prompt exploration	Successful after model update
Final OpenRouter API Integration	Final response generation using DeepSeek Chat	Successful
Prompt Adaptability	Dynamic adjustment based on learner level	Successful

Based on the testing results, the project concluded that the structured answer generator and multi-level explanation system worked successfully. The system was able to generate organized educational responses and adapt the explanation style according to the selected learner level. The refinement process also improved the reliability of the final application by using OpenRouter API and DeepSeek Chat for the final implementation.

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All Tests Passed

Structured Answer Format →  Successful

Beginner-Level Explanation →  Successful

Intermediate-Level Explanation →  Successful

Advanced-Level Explanation →  Successful

Gemini API Integration →  After model fix

OpenRouter API Integration →  Successful

Prompt Adaptability →  Successful

AI Answer Evaluator →  Successful

Interactive Quiz Scoring →  Successful

Basic vs Optimized Comparison →  Successful

Final Demo Scenario

The application will be tested using three different topics, learning levels, study durations, and evaluator roles.

Demo 1 — Beginner

Beginner | 3 Days | Friendly Tutor

Learning Session

Water Cycle

HE Fin Nanyu AI
Your personal AI tutor here to help you learn smarter every day

Basic Prompt Response

Optimized Prompt Response

#1. Definition

The water cycle is nature's way of recycling water. It's the continuous movement of water from the Earth's surface to the sky and back again.

#2. Detailed Explanation

The water cycle has four main steps:

- Evaporation:** The sun heats water in lakes, rivers, and oceans, turning it into water vapor that rises into the air.
- Condensation:** The water cools in the sky and forms tiny water droplets, creating clouds.
- Precipitation:** When the droplets get heavy, they fall as rain, snow, or hail.
- Collection:** Water gathers back in lakes, rivers, and oceans, ready to start the cycle again.

Where does collected water go?

Choose your answer:

- ☒ A. Back to lakes/rocks
- ☐ B. Into space
- ☐ C. Cracks in rocks

Score: 3/5

Nice Effort! Review the explanation again and keep practicing.

AI Answer Evaluator

Write your answer here for AI evaluation:

The water cycle is nature's way of recycling water.

Evaluate My Answer

Rule-Based Answer Evaluation

Accuracy Score: 3/5

The student's answer is factually correct. The water cycle is indeed a natural process that recycles water.

Demo 2 — Intermediate

Intermediate | 7 Days | Friendly Tutor

Learning Session

Newton's First Law

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Your personal AI tutor here to help you learn smarter every day

Basic Prompt Response

Optimized Prompt Response

1. Definition

Newton's First Law, also known as the Law of Inertia, states that an object will remain at rest or in uniform motion in a straight line unless acted upon by an external force.

2. Detailed Explanation

This law introduces the concept of inertia, which is the tendency of an object to resist changes to its state of motion. Inertia depends on the object's mass; the greater the mass, the greater the inertia. For example, a stationary object will not move unless a force (like a push) is applied, and a moving object will not stop unless a force (like friction) acts upon it. This law fundamentally explains why seatbelts are necessary—they prevent passengers from continuing their motion when a car suddenly stops.

3. Real-World Example

Imagine a hockey puck sliding across ice. If the ice is perfectly frictionless, the puck would continue moving at the same speed indefinitely, in a straight line (assuming no other forces eventually slow it down).

4. Summary

Newton's First Law states that objects resist change to their motion unless an external force acts on them.

5. Study Plan

Day 1: Read the definition and explanation of Newton's First Law. Write down key terms (e.g., inertia, external force).
Day 2: Watch a video demonstration of inertia (e.g., a laboratory pulled from under a table). Summarize the video in your own words.
Day 3: Solve 3-5 practice problems involving inertia (e.g., calculating the force needed to move a stationary object).
Day 4: Research real-world applications (e.g., seatbelts, airbags) and write a short paragraph on one application.
Day 5: Review all notes and take the interactive quiz below.

6. Interactive Quiz

Rule-Based Answer Evaluation

Accuracy Score

Score: 3/5

The student correctly identifies Newton's First Law as the "Law of Inertia," which is accurate. However, the answer lacks a proper definition or explanation of what the law actually states.

Completeness Score

Score: 3/5

The answer is incomplete. While it names the law, it fails to describe its key components: (1) an object in motion stays in motion, (2) an object at rest stays at rest, and (3) unless acted upon by an external force.

Clarity Score

Score: 4/5

Demo 3 — Advanced

Advanced | 7 Days | Professional Examiner

Learning Session

Machine Learning

HE Fin Nanyu AI
Your personal AI tutor here to help you learn smarter every day

Basic Prompt Response

Optimized Prompt Response

1. Definition

Machine Learning (ML) is a subset of artificial intelligence (AI) that enables computer systems to learn and improve from experience without explicit programming. It involves algorithms that analyze data, identify patterns, and make decisions or predictions.

2. Detailed Explanation

Machine Learning relies on statistical models and iterative training processes to refine its performance. Key components include:

- Data:** The foundation (e.g., labeled or unlabeled datasets).
- Algorithms:** Mathematical procedures (e.g., decision trees, neural networks).
- Training:** Adjusting model parameters to minimize errors.
- Evaluation:** Testing accuracy with validation data.

Performance Evaluation Results

Answer Technique	Clarity	Structure	Consistency
Standard Prompt	Moderate	Moderate	Moderate
Basic Prompt	Moderate	Low	Moderate
Structured Prompt	High	High	High
Few Shot Prompt	High	Moderate	Very High
Optimized Prompt	Very High	Very High	Very High

Final Analysis

The optimized prompt produced the most educational and well-structured response. Structured prompting improved readability and organization, while few shot prompting improved response consistency.

HE Fin Nanyu AI generated an optimized educational response successfully.

AI Answer Evaluator

Write your answer here for AI evaluation:

Machine Learning (ML) is a subset of artificial intelligence (AI).

Evaluate My Answer

Accuracy Score

Score: 3/5

The answer is incomplete. While it names the law, it fails to describe its key components: (1) an object in motion stays in motion, (2) an object at rest stays at rest, and (3) unless acted upon by an external force.

Demo focus: Basic vs Optimized Response, Study Plan, Interactive Quiz, AI Answer Evaluator, and Role-Based Feedback.

Conclusion



Nazsya AI Tutor was successfully developed as an AI-powered educational assistant using advanced prompt engineering techniques.



The application can generate structured explanations, personalized study plans, interactive quizzes, and AI-based answer evaluation.



The optimized prompt produced clearer, more organized, and more educationally useful responses compared to the basic prompt.



Rubric-based and role-based evaluation helped provide more measurable and meaningful feedback for learners.



Overall, this project shows that prompt engineering can improve the quality, consistency, and usefulness of AI responses in education.